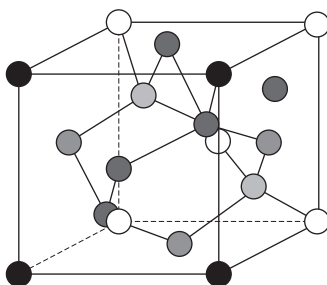


(b) A unit cubic cell of diamond is shown below.



Each side has dimensions of 0.36 nm ($1 \text{ nm} = 10^{-9} \text{ m}$).

The number of carbon atoms in a cubic cell is 8.

The mass of a carbon atom is $2 \times 10^{-26} \text{ kg}$.

Use the information above and the equation:

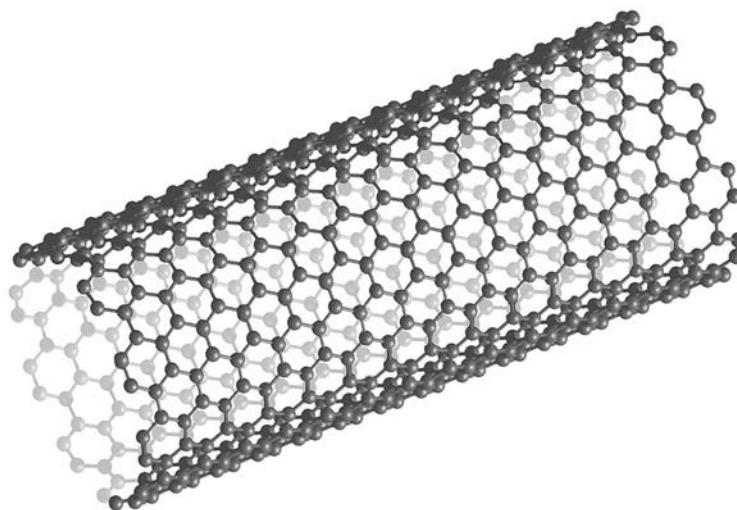
$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of this unit cubic cell of diamond to 2 significant figures.

[4]

Density = kg/m^3

(c) A carbon nanotube is shown below.



Some properties of these nanotubes are compared with three other materials in the table below.

Material	Tensile strength (MPa)	Density (g/cm ³)
stainless steel	505	8.00
aluminium	572	2.81
Kevlar	3 620	1.44
carbon nanotube	62 000	1.34

Explain why carbon nanotubes are likely to replace materials such as steel and aluminium in the manufacture of power lines. [2]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																			
7 Li Lithium 3		9 Be Beryllium 4																	
23 Na Sodium 11		24 Mg Magnesium 12																	
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30						
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89															

Key

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number